TYCUS solutions

COMPANY PROFILE



Heat. Measure. Control.

Industrial heaters, temperature measurement, and process instrumentation

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Lanatan, Balayan, Batangas

About TYCUS

TYCUS Enterprises Corporation, operating as TYCUS Solutions, is an industrial solutions provider based in Balayan, Batangas. Our products are used in chemical processing, power generation, food and beverage, water and utilities, manufacturing, steel and metal, and cement and mining operations across the Philippines.

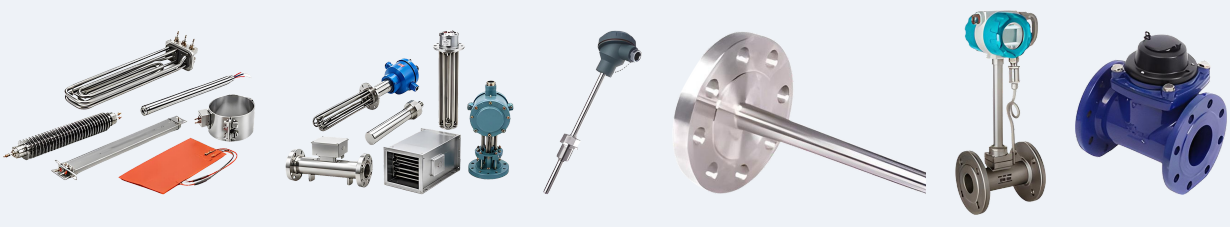
We specialize in industrial and process heaters, custom-built to each customer's required wattage, voltage, dimensions, and operating conditions. Complementing these is our temperature measurement line: thermocouples, RTDs, complete RTD and thermocouple assemblies, thermowells, temperature transmitters, and thermocouple wire.

Together, the two lines let us support the entire thermal loop, from generating heat to measuring and controlling it, through a single supplier.

TYCUS also supplies process valves, flow meters, water meters, pressure transmitters, flow and level switches, electrical control components, and analytical instruments and general laboratory equipment.

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Heat

Industrial and process heaters, built to order

Measure

Thermocouples, RTDs, thermowells, and wire

Control

Transmitters, switches, and process instruments

Our specialized product lines

Heating and temperature measurement belong together. A heater performs only as well as its sizing, its sheath material, and the control loop around it, and that loop starts with the sensor.



Industrial electric heaters and heating elements

Individual heating elements, complete process heaters, and elements for equipment and appliances.

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Temperature measurement products

From thermocouples and RTDs to thermowells, transmitters, and thermocouple wire.

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Each item is built to order by our manufacturing partners. We handle application review, specification, delivery, and local support.

Alongside these

We supply valves, flow, pressure, level, electrical, and analytical and laboratory products from established manufacturers under their own brands, so a project can be sourced from one supplier.



Heating products

The TYCUS heating line covers three groups: individual heating elements, complete process heaters, and elements for equipment and appliances.



Industrial heating elements

Elements for machinery, equipment, and temperature-controlled systems.

Tubular | Finned | Cartridge | Band | Coil | Strip | Flexible | Infrared | Silicon carbide | PTFE



Industrial process heaters

Complete heaters for liquids, tanks, vessels, and air or gas streams.

Flange immersion | Screw plug | Circulation | Duct | Explosion-proof

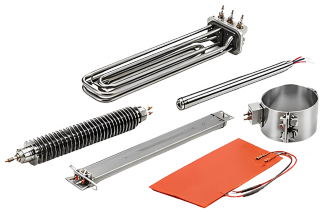


Equipment and appliance heaters

Replacement and OEM elements for ovens, food equipment, and commercial appliances.

Oven elements | Fryer heaters | Beverage machine elements | Washer heaters

Industrial heating elements



Product	What it is	Typical use
Industrial heating elements		
Tubular heaters	Metal-sheathed elements formed to shape. The base component of most industrial heaters.	Ovens, air and liquid heating, platens
Finned heaters	Tubular elements with fins for better heat transfer to air.	Dryers, duct heating, load banks
Cartridge heaters	Cylindrical, high-watt-density elements inserted into drilled holes.	Molds, dies, sealing bars, hot plates
Band and coil heaters	Elements that clamp or wrap around cylindrical parts.	Extruder barrels, nozzles, hot runners
Strip heaters	Flat elements for surface and air heating.	Platens, panels, enclosures
Flexible heaters	Thin heating films and mats that follow the shape of the surface.	Enclosures, small tanks, equipment needing even surface heat
Infrared heaters	Radiant elements that heat the product without contact.	Drying, curing, thermoforming
Silicon carbide elements	High-temperature elements for furnace service.	Industrial furnaces, kilns
PTFE (Teflon) heaters	Corrosion-resistant immersion heaters.	Plating lines, acid and chemical baths

Process, equipment, and appliance heaters



Product	What it is	Typical use
Industrial process heaters		
Flange immersion heaters	Tubular elements mounted on a flange and installed directly in the liquid.	Tanks, vessels, large liquid volumes
Screw plug heaters	Immersion elements on a threaded plug for smaller openings.	Small tanks, reservoirs, oil and water heating
Circulation heaters	An immersion heater inside a vessel that heats fluid as it flows through.	Circulation loops, oil, water, and gas streams
Duct heaters	Element banks mounted in an air duct.	Process air, drying, ventilation
Explosion-proof heaters	Heaters with certified enclosures for classified locations.	Oil, gas, and chemical service
Equipment and appliance heaters		
Oven heating elements	Formed tubular elements made to the oven's cavity and wattage.	Baking, drying, and curing ovens
Fryer heaters	Immersion elements shaped for fryer tanks.	Food processing and commercial kitchens
Beverage machine elements	Compact elements and cast heaters for water and steam.	Coffee and beverage equipment
Washer heaters	Immersion elements for wash water.	Washing and cleaning equipment

Heater designs outside this list can be reviewed on request.

Heating solutions by application

The right heater depends on what is being heated, the temperature, the power available, and how the heater is installed. These are the applications we specify for most often.



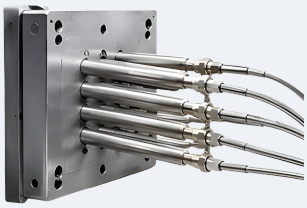
Liquid heating

Water, oil, chemicals, and other process fluids in tanks, vessels, and circulation loops.



Air and gas heating

Process air, drying, ventilation, and gas streams.



Surface heating

Molds, dies, pipes, barrels, plates, and other solid surfaces.



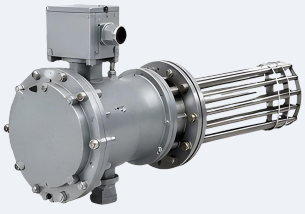
Industrial ovens

Elements for ovens, dryers, heating chambers, and curing lines.



High-temperature heating

Furnaces, kilns, and other high-temperature thermal processing.



Hazardous areas

Explosion-proof heaters for classified locations in oil, gas, and chemical service.

Temperature measurement products

Sensors, protection, and wiring for the measuring side of the thermal loop, built to the sensor type, sheath, insertion length, and process connection each application needs.



Thermocouples

Armored and wear-resistant assemblies in types K, J, T, E, and N for general process service.



High-temperature thermocouples

Types S, R, and B with ceramic protection tubes for furnaces, kilns, and incinerators.



RTDs

Resistance temperature detectors for accurate, stable measurement at low to moderate temperatures.



Thermocouple wire

Thermocouple and extension wire by type and gauge, with insulation and overbraid to suit the temperature.



Flanged, welded, and coated thermowells

Wells for higher pressures and flanged nozzles, with special coatings for corrosive media.



Threaded thermowells

Screw-in wells that protect the sensor and allow it to be removed without draining the process.

Sensors, transmitters, and accessories



Product	What it is	Typical use
Industrial thermocouples		
Armored thermocouples	Mineral-insulated, metal-sheathed thermocouples that can be bent to route and resist vibration.	Pipes, vessels, ovens, machinery
High-temperature thermocouples	Noble-metal thermocouples in ceramic or alloy protection tubes.	Furnaces, kilns, heat treatment
Wear-resistant thermocouples	Assemblies with hardened protection tubes for abrasive flows.	Boilers, cement, powder and bulk solids handling
Incinerator thermocouples	Assemblies built to withstand the abrasion and corrosion of waste combustion.	Waste-to-energy plant incinerators, flue gas
RTDs, transmitters, and accessories		
Resistance temperature detectors	Platinum resistance sensors such as Pt100, in 2-, 3-, or 4-wire connection.	Food and beverage, water treatment, tanks and pipelines
Temperature transmitters	Convert a thermocouple or RTD signal to a standard output such as 4 to 20 mA for the control system.	Long cable runs, PLC and DCS inputs
Thermowells	Threaded, flanged, welded, and special-coated wells machined to the insertion length and process connection.	Pressurized lines and vessels, corrosive media
Thermocouple wire	Thermocouple-grade and extension wire supplied by type, gauge, and insulation.	Sensor fabrication, wiring from sensor to instrument

Thermocouple types

Type	Conductors	–200 °C **Typical range** 1,800 °C	Typical use
K	Nickel-chromium / nickel-aluminum	**–200 to 1,260 °C**	General purpose; ovens, furnaces, heaters
J	Iron / constantan	**0 to 760 °C**	Plastics machinery, older equipment
T	Copper / constantan	**–200 to 370 °C**	Low temperatures, food, refrigeration
E	Nickel-chromium / constantan	**–200 to 900 °C**	High signal output at low to mid temperatures
N	Nicrosil / nisil	**–200 to 1,300 °C**	More stable alternative to type K at high temperature
S	Platinum-10% rhodium / platinum	**0 to 1,600 °C**	High-temperature processes, reference measurement
R	Platinum-13% rhodium / platinum	**0 to 1,600 °C**	High-temperature industrial furnaces
B	Platinum-30% rhodium / platinum-6% rhodium	**0 to 1,700 °C**	Very high-temperature furnaces and kilns

Ranges are typical for each type. The working limit of a specific sensor depends on its sheath, wire size, and construction, and is confirmed in the quotation.



Other products we supply

In addition to our own heating and temperature measurement lines, TYCUS supplies the following products from established, reputable manufacturers. They are offered under the manufacturers' own brands, with the make and model selected to suit each client's application and budget.



Product	What we supply
Process flow meters	Electromagnetic, turbine, positive displacement, ultrasonic, and Coriolis meters for liquids and gases, selected by fluid, line size, and accuracy required.
Water meters	Meters for plant, utility, and building water supply, for consumption monitoring and billing.
Pressure, flow, and level instruments	Pressure transmitters, flow switches, and level switches for process monitoring, alarms, and pump protection.
Electrical control components	Contactors, overload relays, circuit breakers, fuses, motor starters, control relays, and pilot devices for control panels.
Analytical and laboratory	Analytical instruments and general laboratory equipment for process, water, and quality control testing.

Process valves

TYCUS supplies control, safety, and tank protection valves from established manufacturers. They are offered under the manufacturers' own brands, with the size, pressure rating, and materials selected to suit each application.



Product	What it is	Typical use
Control valves and regulators		
Globe control valves	Pneumatic or electric actuated valves that throttle flow to a control signal.	Flow, pressure, level, and temperature loops
Regulators	Self-acting pressure and temperature regulators that need no external power.	Steam heating, pressure reduction stations
Safety and protection valves		
Safety valves	Spring-loaded valves that open at a set pressure to relieve overpressure.	Boilers, pressure vessels, compressors
Breather valves	Pressure and vacuum relief valves that protect storage tanks and limit vapor loss.	Fuel, chemical, and solvent storage tanks
Fusible link valves	Valves held open by a fusible link that shut automatically in a fire.	Fuel oil and gas lines, tank outlets

Brands we supply

Complementing our core lines, TYCUS supplies proven products from established manufacturers.

YOKOGAWA | ROSEMOUNT | Endress+Hauser

ABB | SIEMENS | Schneider Electric

Danfoss | UE UNITED ELECTRIC CONTROLS | FlowT

Skyray Instrument | DRAWELL Artist of Science | BAITE

鑫泉 XINQUAN | SUNOCE日洋 | WTsensor

SDWM Fluid Technology | ZNC | THINKTANK®

Make and model selected to suit each application and budget.

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Industries we serve

Chemical processing	Reactor and vessel heating, thermal oil and heat transfer fluid heating, corrosive liquid and bath heating, and temperature maintenance for tanks and pipelines.
Food and beverage	Ovens and dryers, fryers, wash and process water heating, sealing and packaging equipment, and RTDs for process and storage monitoring.
Power generation	Fuel oil and lube oil heating for turbines and generators, boiler feedwater preheating, hopper and duct heating, and anti-condensation heaters for motors and panels.
Water and utilities	Water and chemical tank heating, anti-condensation heating for panels and enclosures, incinerator and boiler thermocouples, and RTDs with transmitters.
Manufacturing	Molding and extrusion, industrial ovens and furnaces, curing and drying lines, and replacement elements, thermocouples, and thermocouple wire for existing equipment.
Steel and metal	Furnace and heat treatment heating, plating and pickling bath heating, die and mold preheating, and quench oil heating.
Cement and mining	Heavy fuel oil preheating, hopper and chute heating, lube oil heating for mills and crushers, and freeze protection for slurry lines.

Manufacturing and quality

TYCUS products are built to order by our manufacturing partners, each a specialist in its own line: one in electric heating elements, one in temperature sensors. TYCUS sets the specification with you and checks the technical drawing before anything is built.

1

Application review

We look at what is being heated or measured, the temperature, the power available, and how it is installed.

2

Specification

We set the specification with you and check the technical drawing before anything is built.

3

Built to order

Our manufacturing partners build each item to the agreed specification.

4

Delivery and support

We handle delivery and stay on as your local point of support.

Heaters	
Custom work	Size, shape, power, voltage, sheath material, and mounting to specification.
Sheath materials	Stainless steel 316L, Incoloy, and other materials to suit the medium.
Flange standards	ANSI, DIN, and JIS.
Factory testing	Dielectric testing on every unit; power tolerance of ±2%.
Compliance	CE, UL/cUL, RoHS, and REACH documentation available depending on the product, with ATEX and IECEx-certified options for hazardous-area heaters.

Temperature sensors	
Custom work	Sensor type, sheath material and diameter, insertion length, process connection, and head or cable termination to specification.
Thermowells	Precision-machined to the specified insertion length, process connection, and material, with protective coatings available for corrosive or abrasive media.
Calibration	Calibration certificates available through our ISO/IEC 17025:2017 accredited partner laboratory.

TYCUS solutions

Send us your requirements.

For all product inquiries, send us your requirements and we will respond with our recommendation and price quotation.

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